COMP3710 OASIS Brain Analysis Feasibility Review

Current implemented module: baseline versus improved 2D U-Net tissue segmentation

Dataset audit	Geometry	Labels
11,328 pairs; 354 subjects	256 x 256 for every pair	0/85/170/255 mapped to 0-3
Split protocol	Smoke test	Consultation status
Seed 42; subject-disjoint	Baseline, 1 epoch, 8 cases/split	Tutor check-off not yet recorded

Purpose and decision requested

This prototype segments preprocessed grayscale OASIS MRI slices into four tissue classes for human review. I request confirmation that the Easy-difficulty scope, the subject-disjoint comparison, and the stated acceptance criteria are appropriate before full training.

Method and acceptance criteria

Both models will use the same saved seed-42 subject-level manifest. The baseline is a plain 2D U-Net; the proposed model adds residual encoding, attention-gated skips and bottleneck dropout. Acceptance evidence: no subject overlap; four-class logits; class-wise held-out Dice/IoU; parameter, VRAM and inference records; and 3-5 ranked FP/FN failure panels. The course target is Dice >= 0.90 per tissue label.

Verified local evidence

Read-only audit of the local source found 11,328 aligned image/mask pairs from 354 subjects. All pairs are 256 x 256. Raw mask pixel counts are 0: 536,466,114, 85: 41,266,731, 170: 85,030,267, 255: 79,628,696; all map to classes 0-3. Four deterministic audit samples were saved. Unit tests cover nested pairing, subject allocation, encoded-label remapping, and the audit manifest.

Smoke run: evidence of execution, not model quality

A local GPU baseline smoke run recorded PyTorch 2.11.0+cu128 on cuda: one epoch, 8 held-out cases, 1,927,940 trainable parameters, 0.088 s inference, and 284.6 MB peak VRAM. Held-out mean Dice = 0.0182; mean IoU = 0.0094. These deliberately small-run values validate the pipeline and failure-reporting route only; they do not support a performance claim.

Risks, budget and fallback

Risks: label imbalance, memory demand, thin tissue boundaries and a high mean masking class failures. Budget: train baseline and improved models on the identical full manifest; preserve checkpoints, learning curves, JSON metrics and 3-5 failure panels. Fallback: retain the simpler baseline if the improved model does not show a class-wise and qualitative benefit at acceptable resource cost.

Questions for tutor confirmation

1. Confirm the Easy scope and the four-class 2D OASIS target. 2. Confirm Dice >= 0.90 per label as the evaluation target and whether background is reported separately. 3. Confirm the required training budget and acceptable evidence for the midpoint record.